

Assignment 9

Title: Case Study

Aim:

By using concepts of OOP, develop a banking solution containing following operations
1) Create an account 2) Deposit money 3) Withdraw money 4) Honor daily withdrawal limit 5) Check the balance 6) Display account information

Concepts of OOP

i. Class and Object

1. A class is a blueprint for creating objects and objects are instances of classes.
2. In the program, classes like Account, Bank, Customer, Savings Account define the structure.

ii. Inheritance

1. Inheritance is a mechanism in which one object acquires all the properties and behaviours of a parent object.
2. Types of inheritance
 - a) Single inheritance
 - b) Multiple inheritance
 - c) Multilevel inheritance
 - d) Hierarchical
 - e) Hybrid inheritance
3. In the program, class Bank extends class Account, inheriting its attributes and methods. Thus, single inheritance is used.



iii. Abstraction

1. Abstraction is the process of hiding certain details and showing only essential information to the user.

2. Abstraction can be achieved with abstract classes or interfaces.

a. Abstract class: It is a restricted class that cannot be used to create objects. To use it, it must be inherited from another class.

b. Abstract method: It can be only used in an abstract class and it does not have a body.

3. In the program, the class Account is abstract, providing blueprint for other classes.

iv. Encapsulation

1. Encapsulation refers to bundling of data and methods that operate on that data within a single unit.

2. It can be achieved by declaring instance variables of a class as private, which means they can be accessed only within the class.

3. Getters and setters are defined to allow the access outside to instance variables.

4. In the program, instance variables like balance and account type are encapsulated within Account class and methods control access to these variables.

Polymorphism

Polymorphism

Single action

ways.

It is of two

Runtime polymorphism

Dynamic polymorphism

In the program

through methods

where methods

and display

Constructors

1. Constructors

initialize

when object

is created: Constructors

a) Non-parameterized

b) Parameterized

c) Copy constructor

with an

existing object

available

object

Constructors

Account

Polymorphism

Polymorphism means having many forms. Single action is performed in different ways.

It is of two types:

Runtime polymorphism: method overloading

Dynamic polymorphism: method overriding

In the program, polymorphism is achieved through method overriding in Bank class where methods like withdraw, deposit and display are implemented differently.

Constructors

Constructor is a special method used to initialize objects. The constructor is called when object is created.

Constructors are of ~~two~~ ^{three} types

a) Nonparameterized/default: constructor with no parameters

b) Parameterized: constructor consisting of parameters

c) Copy constructor: constructor which is passed with another object which copies the data available from passed object to newly created object.

Constructors are used in classes like Account, Bank and Customer.